Did easier prime-order certification select the published curves?

Date: 2026-10-23 UTC. **Status:** Complete exact factorization of n−1 and n+1 for all nineteen same-field published and omitted curves. This tests one possible 1997 workflow proxy; it does not reproduce the original program.

Result

No exact separator appears. For every one of the nineteen prime group orders n, the product F of *all* factors of n−1 at or below one million is smaller than √n. Thus a Pocklington proof using only these small factors is unavailable for **every** published and omitted curve. Complete n−1 factorizations exist in the sealed run, but they require at least one larger proven factor to cross the theorem's size condition.

The size of this small-factor product is mixed by label. In the 239-bit field and P-256, published curves mostly have more small factors of n−1; across all four fields, the published value is larger in **19 of 25** within-field published-versus-omitted comparisons. Those pairs share curves and are not independent trials. The pattern fails as a universal selector: omitted P-224 block 1485 has a **53-bit** small-factor product versus the published P-224′s **29-bit** product, and published prime192v3 has only **3 bits** versus **9** and **17 bits** for the two omitted 192-bit curves. The negative-control n+1 comparison also favors published curves in **16 of 25** pairs.

Field	Omitted n−1 small-factor product bits ($B=10^6$)	Published controls, same metric
192-bit	blocks 2198: 17; 5405: 9	prime192v1: 18; prime192v2: 16; prime192v3: 3
239-bit	blocks 36: 27; 1562: 21; 2491: 20; 2682: 11; 2863: 5	prime239v1: 50; prime239v2: 23; prime239v3: 24
P-224	blocks 777: 28; 1485: 53; 5736: 21	secp224r1: 29
P-256	block 876: 15	prime256v1: 69

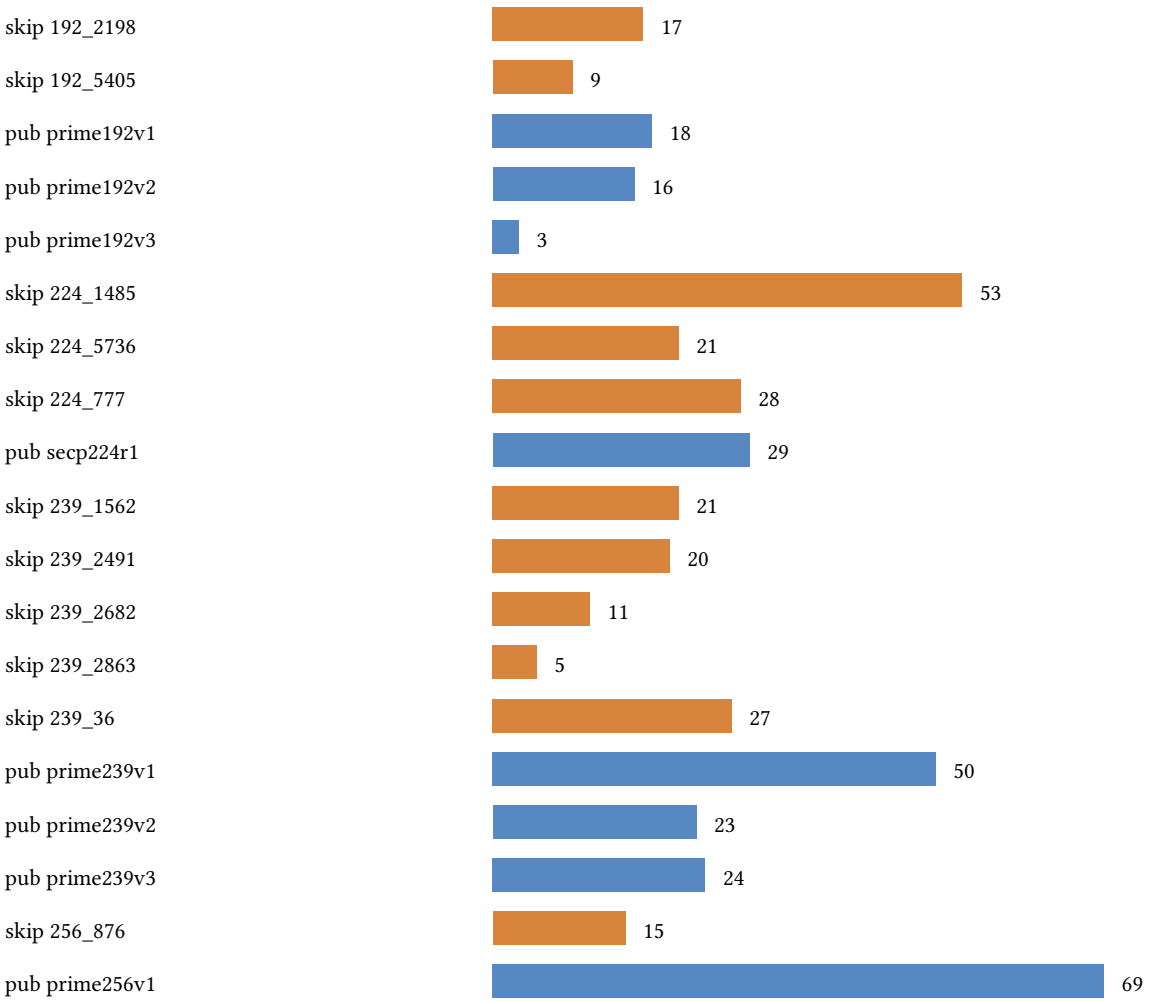


Figure 1: Small-factor products of n minus one

Figure 1. Exact bit lengths of products of prime powers with prime base at most one million; orange marks omitted candidates, blue published controls. None crosses √n. Editable source: analyze.py.

Method and checks

The pre-run protocol fixed the 11 omitted classes, eight same-field published controls, both neighbors `n−1` and `n+1`, four factor bounds and Pocklington-size metrics. It pinned the archived input digests and a 180-second per-case watchdog. The resulting **38 real factorizations** and three arithmetic controls (`15`, `60`, `101`) all completed. For every case, Sage verification matched the raw GP transcript and source curve UID, checked the full product, and checked each factor with proof-enabled primality. Sage separately checked the GP outputs but may share arithmetic internals; this is not a second factorization. Two injected launcher failures returned nonzero with no completion marker (control receipts).

The run started from clean commit `895cfe6d51e22e44649272f2da1aa263525e68c0` on macOS arm64 with Python 3.12.8, GP/PARI 2.17.3 and SageMath 10.10.rc0 (environment). The 41 serial GP calls took **62.948 seconds** in aggregate; the longest took **20.695 seconds**, and the maximum sampled resident size was **34,488,320 bytes**. These are measurements of the modern local factorization, not a model of 1997 certificate effort. The raw external run was copied byte-for-byte into this write-once directory. The derived analysis contains every row, every bound, exact pair counts and the source verification hash.

What this does and does not say

Pocklington's theorem gives a deterministic primality certificate when a completely factored divisor `F` of `n−1` exceeds `√n` and suitable modular witnesses are supplied. That was a known route in the 1990s (Morain's 1997–98 account, `retrieved`, primary).

We tested the *size condition* and factor structures, not the modular witnesses, recursive proof effort or the original software. The 1998 ANSI X9.62 standard specifies a probabilistic primality test in Annex A.2.1; a correct probable-prime test does not require factoring `n−1` and cannot reject a truly prime `n` as composite. A historical implementation error, timeout or additional proof policy remains possible but is not evidenced by this calculation.

Even with complete `n−1` factors, "easier to prove" needs a specified algorithm. Taking the largest prime-power components first, the minimum number needed to exceed √`n` is either one or two across these nineteen curves. This metric is also mixed: omitted P-256 block 876 needs one large component whereas published P-256 needs two, although the published curve has a larger product of small factors. The different proxies rank this pair in opposite directions. None is a curve-group weakness. Factors of `n−1` are **not** factors of the prime group order `n`, so they do not create a Pohlig–Hellman shortcut. An auxiliary-input DLP model would need its own explicit oracle and attack analysis.

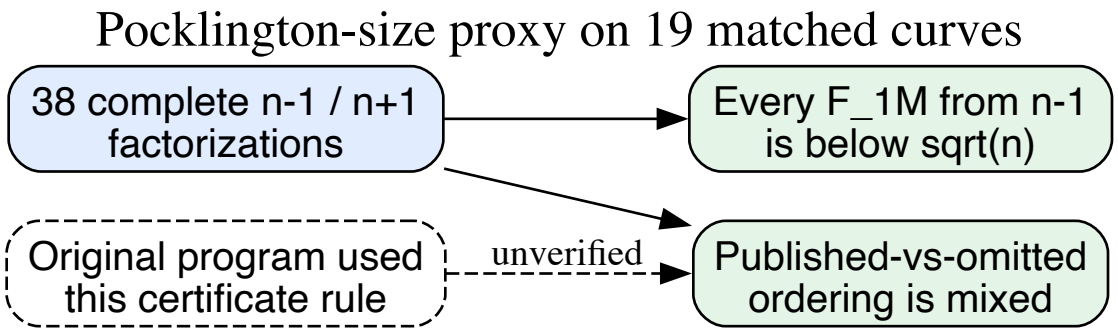


Figure 2: Inference boundary

Figure 2. Solid arrows are exact computations and bounded inferences. The dashed historical selection premise remains unverified. Editable source: inference.dot.

Under the user's working premise that all eleven candidates were reached and passed over, this study excludes the simple rule "reject when the prime order lacks enough small `n−1` factors for a Pocklington certificate" at every predeclared bound through one million: **all nineteen** lack enough. It also rules out a global monotone threshold on this small-factor product, because same-field published and omitted values overlap in both directions. It does not identify what the original program checked or any common cryptanalytic weakness. The selection-rule problem remains open; the canonical ECDLP graphs were checked and gain no attack edge from this workflow proxy.

Reproduction

```
python3 research/nist_seed_order_certificate_20261023/run.py --out NEW_WRITE_ONCE_DIRECTORY
python3 research/nist_seed_order_certificate_20261023/analyze.py
```

The archived directory has exact commands, raw stdout/stderr, factors, versions, resource receipts, Sage checks and a completion marker. Source labels: `internal` for pinned curve and earlier selection records; `measured` for this local run; `separately checked` for Sage factor verification; `derived` for the bounded size and pair comparisons. No official hypothesis status changes here.